

Release agents can be applied using a spray canister or by using the compactor's tire spray system.

In some areas, natural vegetable oil is substituted for water in the spray system reservoir. Do not use petroleum distillates as they are harmful to asphalt and the environment.

- Before the start of the paving and compaction process, operate the pneumatic compactor on a paved surface behind the starting point. Operate at high speed to build up heat in the rubber tires by flexing them.
- When using a release agent, wet the tires thoroughly just as the compactor is ready to start its first pattern.
- Check the temperature of the asphalt layer and guide the pneumatic compactor operator into the correct temperature zone.
- Be alert for any sign of excessive asphalt pick-up on the tires. Especially watch for asphalt clumps falling off the tires when the compactor stops and reverses.

To prepare the pneumatic compactor for placement in the desired position behind the paver, plan your approach to heating the tires. Apply a release agent (if needed), and keep the tires hot.

- If you see excessive asphalt pick-up, immediately clean the tires. Move the compactor back to a cooler temperature zone. Gradually move the compactor forward allowing the tires to heat up prior to reaching the desired temperature zone.
- Once the tires are heated, keep them heated. If there is an interruption to the paving and compaction process, do not park the pneumatic compactor. Move the compactor to a place on the asphalt layer where it can continue to roll in order to keep the rubber tires heated.

[DEEP PNEUMATIC TIRE MARKS]

Using a pneumatic compactor on hot asphalt layers, especially layers 75 mm (3") thick or thicker can result in deep tire marks that are difficult to clean up, particularly behind vibratory screeds.

Ordinarily, a pneumatic compactor is used during the intermediate phase of compaction on an asphalt layer that is already close to final target density. The tire marks it leaves in the mat are normally shallow and can be smoothed out by the finish-phase compactor.

However, if the compactor is used during the initial compaction phase or if the pneumatic compactor rolls an area where the mat is thicker and hotter than normal, the rubber tires can leave deep marks that do not clean up easily during finish compaction.

Using a pneumatic compactor during the initial phase is usually done when compacting a base or binder layer that is going to have another layer laid on top of it. In that instance, the tire marks and loss of smoothness are less of an issue.

Using a pneumatic compactor during the initial phase on the final layer (wearing layer) of asphalt is not common because the final layer is often measured for smoothness. A pneumatic compactor is normally in the intermediate position when compacting the final asphalt layer.

If deep tire marks appear during the compaction of the final layer:

- Move the pneumatic compactor farther behind the paver where the asphalt layer is cooler, or
- Decrease tire pressure to flatten the tires somewhat and reduce the tire contact pressure.



Tire marks left by the pneumatic compactor can normally be cleaned up by the finish-phase compactor.



Deep tire marks created by operating on a thick layer composed of a tender mix will be difficult to clean up.

[VIBRATORY DRUM IMPACT MARKS]

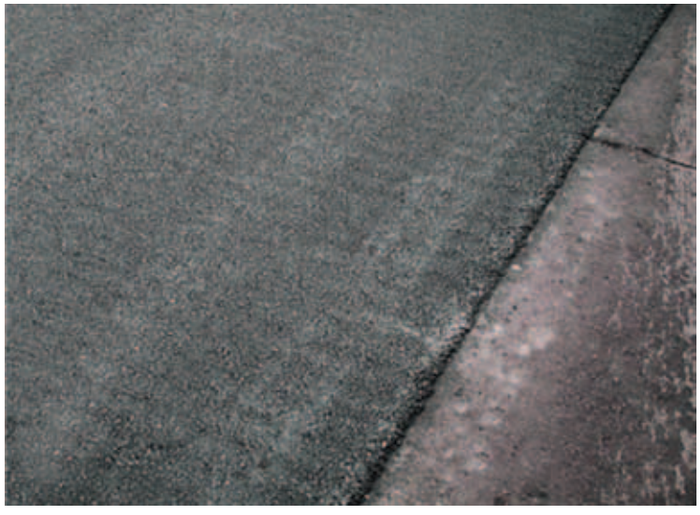
When too much vibratory compaction energy is applied to an asphalt layer, impact marks that do not clean up during the finish phase may appear on the surface of the asphalt layer.

In Unit 2, "Forces of Compaction," you learned about the affects of weight and amplitude. In Unit 3, you learned about other factors, like working speed and layer thickness, that influence the compaction process. In the lower-left photo, it is obvious why the asphalt layer has drum impact marks.

The compactor made numerous, slow vibratory passes over the joint between the asphalt layer and the concrete gutter in an effort to knock down the asphalt layer for a better height match. It is certain that the drums were bouncing in this area. You can even see the resulting white, powdery surface that indicates fractured aggregate. In this case, the problem was created by the paving crew that paved the joint height too high. The compactor can only reduce the thickness of the layer so much. When the layer becomes dense, the drums will begin to bounce and leave impact marks.

By way of review, if you begin to feel drum bouncing or if you begin to see impact marks in the surface of the asphalt layer, you should adjust one or more of the following variables:

- Check the working speed to make sure you are operating in the range that produces 26 to 46 impacts per meter (8-14 impacts per foot).
- Switch to a lower amplitude setting.
- If available on the machine, switch to a higher frequency.
- Operate with one drum vibrating and one drum static.
- Operate in the static mode.





Drum impact marks, above. At right, establish a pattern that does not distort the hot asphalt when reversing.

[REVERSING ON THE ASPHALT LAYER]

Most of the rolling patterns used by steel drum, vibratory compactors require the compactor to stop and reverse on the hot asphalt layer. Unit 5 covers rolling patterns in detail. In this unit, we review the potentially significant problems caused by stopping on a fresh asphalt layer.

A steel drum compactor always stops at an angle on the fresh asphalt layer when reversing at the end of a pass. Leaving a stop mark that is at least a 30° angle will make it easier for the next compactor to clean up the stop mark. Be sure to use a compactor with drum width that makes it possible to complete an angled turn while stopping to reverse.

The operator will have a hard time stopping and reversing on a relatively narrow mat when using a compactor with drums that are 200 cm (79") wide or wider. Prior to the start of a project, be sure to understand all the paving widths that the compaction crew will encounter. Use a tool like the Cat Interactive Production Calculator to match the compactor's production to the paver's production.

As a general rule, select compactors with drum widths that cover the mat in three overlapping passes. These compactors will have more space to make angled stops on the fresh asphalt layer. When production requires you to select compactors with wider drums that cover a mat in two overlapping passes, be sure to help the compactor operators establish a pattern that does not distort the hot asphalt.



[PARKING ON THE ASPHALT LAYER]

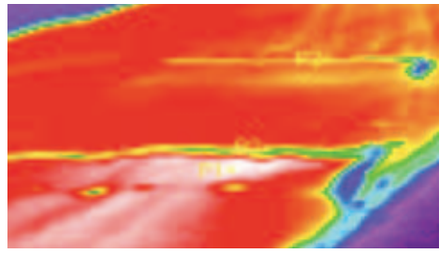
An asphalt compactor of any type should never be stopped to wait on an asphalt layer until that layer is completely compacted and has cooled below 20° C (70° F). It is especially important to avoid stopping and waiting on an asphalt layer that will be measured for smoothness. Make every effort to stop the compactor in a location that will not damage the freshly placed mat.



Stopped and parked partially on shoulder.



Pulled away after six-minute stop.



Thermal image shows cold areas.

Anytime a roller stops and parks on a fresh asphalt layer, the drums or tires will dent the mat. The photos illustrate the effect of a six-minute stop on the asphalt layer. In this example, the paver had stopped while waiting for more haul units. The initial phase compactor completed its pattern and the operator parked with the drums at an angle and with a portion of the drums on the 90 cm (3') emergency lane and a portion of the drums extending onto the driving lane that is going to be measured for smoothness.

After a six-minute stop, the paving process started again and the compactor operator started a new pattern. A view of the mat where the compactor had been parked reveals that water has pooled in the area where the drums sat and where the mat has been dented.

A thermal image shows that the steel drums caused significant heat loss in the mat where the drums sat. The temperature of the mat is 65° C (150° F) where the two steel drums were located. The mat has two more compaction phases remaining, intermediate and finish. However, the temperature of the mat in this area is now much lower than found in the normal patterns.

At the end of the shift, the finish-compacted driving lane was measured for smoothness using a profilograph. The Profile Index at the location where the compactor parked for six minutes shows two depressions and bulges caused by the steel drums and not cleaned up during the intermediate and finish compaction phases. The Profile Index is proof that parking on a hot mat for any length of time and at any angle is likely to leave permanent bumps in the mat.

If the paving process is interrupted, Caterpillar recommends parking any affected compactor on a surface that is cold and completely dense or that is not part of a driving lane. If no suitable parking area is available, the compactor(s) should deactivate the vibratory system, move to an area of the asphalt layer that is away from an active rolling pattern and continue to operate at slow speed on the mat until the paving process resumes.

[STOPPING FOR WATER REFILL]

Depending on the size of the water spray system reservoirs and the climate conditions, a steel drum compactor must stop for a water refill one or more times per shift. It is important that the crew plan for water stops to avoid long interruptions to the compaction process and to avoid stopping the compactor on any portion of the asphalt layer that is part of the driving lane. Following are some recommendations for minimizing disruptions during water refill stops.

Park the compactor on a cold, compacted surface or on the shoulder when stopping to refill water spray system reservoirs. Often this means that the water supply truck must have a long hose. In some situations, the water supply truck must park on a cold surface and the water hose must extend across the 3.65 m (12') wide driving lane to reach a compactor parked on the opposite shoulder. Plan ahead by knowing the maximum hose length that will be required for water refills on each project. Make sure the water supply hose is long enough for every situation.

On some projects, the mat will have two unconfined edges without a shoulder for the compactor to stop on. Or, the mat may have one unconfined edge and no room on the opposite side for the compactor to get off the mat. In those situations, plan to provide sturdy boards to place along the unconfined edge of the mat. The compactor can exit and enter the mat by using the boards for support and not crush the unconfined edge of the mat.

Minimizing the length of the water refill stops is important. The paving process may have to be temporarily halted until the compactor(s) can resume the pattern(s). Or, depending on the project and the types of compactors, you can move the intermediate compactor up to the initial phase while the initial phase compactor is stopped for water refill.

No matter what steps you take during the water refill process, there is a likelihood that the temperature of the mat will vary and it is possible that density will vary according to the severity of temperature loss. One way to save time during water refill stops is to position the water supply truck in a convenient location for quickly refilling the compactors.





Compactor parked on the shoulder during water refill (left) to avoid denting the mat in the driving lane. Provide a way for the compactor to get off a mat with unconfined edges, if necessary.

COMPACTION CALCULATOR

Trucking
Paver Speed
Compaction
Windrow
Yield
Slope
Thickness
Job Summary
Legal

General Inputs

Paving Thickness:	[2.95] in	[75.0] mm
Paving Width:	[12.00] feet	[3.658] meter
Material Density Uncompacted:	[130] lbs/ft³	[2082] kg/m³
Truck Capacity or Total Tonnage:	[881.8] ton	[800.0] tonnes
Length of Mat at 100% Yield:	[4598.70] feet	[1402] meter
Actual Length of Mat produced:	[4691.60] feet	[1430] meter

% Yield for given truck load or tonnage: [102]

Thickness:	[2.95] in	[75] mm
Length of Mat produced:	[4691.6] feet	[1430] meters
Width:	[12] feet	[3.658] meters

Exit

Using the "Yield" feature on the Cat Interactive Production Calculator can help you position the water refill truck.

An example using the Cat Interactive Production Calculator demonstrates the importance of planning for positioning of the water refill truck. On this project, the production rate is 200 tonnes (220 tons) per hour. The paving width is 3.65 m (12'). The paving thickness is 75 mm (3"). The weight of the mix laid down by the screed is 2082 kg/cm^3 (130 lbs/ft^3). One lane of a highway project is being paved continuously for the entire shift.

Based on the water reservoir capacity and the climate conditions, the compactor must stop for a water refill every four hours. In four hours, the paver should have laid down 800 tonnes (880 tons). Once all the data is entered into the yield calculator, the result is 1402 m (4598') of paving in four hours. You should look for a location on the project that is close to the calculated yield distance, that is a convenient place to park the water supply truck, and that is appropriate for the compactor to stop. If there is no place for the compactor to park without denting the mat, plan to get the compactor off the mat using boards.

Having the water supply truck in position and personnel ready to refill the water spray system reservoirs will shorten the interruption to production and minimize mat temperature variations.

[TIGHT RADIUS COMPACTION]

In some applications, most notably streets and parking lots, there will be areas where the compactor will be working on curved surfaces such as cul-de-sacs and around dividers and other obstructions. The correct equipment and the correct techniques must be used to avoid distorting the fresh asphalt layer when compacting curved surfaces.



Conventional 170 cm (67") drum compactor rolled continuously around the radius.



The outer edge of the drum stretched and distorted the fresh asphalt.

Anytime the mat has a tight radius, the mat will be distorted if a conventional, high-production, steel drum compactor makes a continuous pass around the radius. The outer edge of the drum is covering a longer distance than the inner edge of the drum. Consequently, the outer edge stretches the mat in an effort to turn the radius at the same speed as the inner edge.

If the only available compactor is a conventional, high-production model, then a unique rolling pattern must be used to avoid distorting the mat.

The operator should work toward the radius along one edge of the mat. Roll forward (1) into the radius, stopping at an angle close to the outer edge of the radius. Reverse back along the same path.

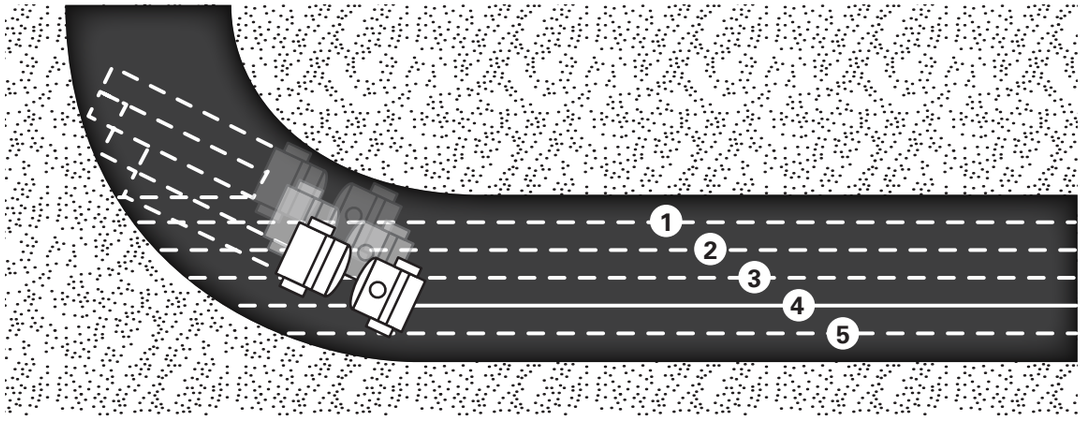
Move toward the center of the mat (2) overlapping the first pass slightly and roll forward into the radius, again stopping at an angle close to the outer edge. Reverse back along the same path.

Repeat this pattern as many times as necessary (3,4,5) to cover the width of the mat.

Reposition the compactor at the start of the radius so the compactor is at an angle to the passes already completed in the radius area. Roll forward and backward to finish the remainder of the radius area.

Then, reposition the compactor to begin straight passes going forward out of the middle of the radius.

BENDS OR JUNCTIONS



Rolling pattern for radius compaction using conventional steel drum compactor.



Utility-size compactors make it easier to compact around obstacles without tearing the fresh asphalt.



Split drum compactors can perform radius compaction without tearing fresh asphalt.

On low-production jobs, another solution is using a utility-size compactor for radius compaction.

Utility class compactors with drums less than 1 m (40") wide can turn sharper on a fresh mat without tearing the mat. For cul-de-sacs, city streets, parking lots and other lower-production projects, a utility-size compactor provides versatility and usually enough productive capability to keep up with the paving process.

Another option in some areas is the split drum asphalt compactor. These units have unique propel systems which synchronize steering and drum speed to move one half of the drum at a faster speed than the other half. When the steering wheel is turned, the outer half of the drum (along the larger arc of the radius) speeds up in comparison to the inner drum half (along the smaller arc of the radius). The more the steering wheel is turned, the greater is the difference in the speeds of the two drum halves. Thus, the drum halves cover different distances but in the same time frame. The fresh mat reaches target density without distortion.

If split drum models are distributed in your area and you work on projects with radius compaction, Caterpillar recommends including one or more split drum models in your compactor fleet.

User Tip: *This radius pattern increases the number of passes and requires carefully positioning the compactor. All this movement takes time and means that the paver is likely to move so far ahead that the mat will lose too much heat before the initial phase compactor can catch up. Caterpillar recommends that the paver make several short stops, three minutes for example, as it completes paving the radius. Short paver stops will not seriously affect mat temperature and will enable the initial compactor to keep pace with the paver.*

[INCONSISTENT DENSITY]

Many public works departments now require not only high asphalt layer density, but also consistent density. There may be pay factors associated with standard deviation derived from multiple core measurements or percent within engineering limits derived from multiple core measurements.

One of the responsibilities of the paving crew is to present a uniform asphalt layer to the initial phase compactor. Behind the paver, to the extent possible, the mat should be:

- Uniform screed laid density
- Uniform thickness
- Uniform temperature



Consistent operation of the paver and the initial phase compactor are key to achieving consistent density.

User Tip: *Caterpillar recommends periodic checks of screed-laid density across the width of the mat and checks of surface temperature across the width of the mat. The public works department may have written specifications for uniformity of screed-laid density and mat surface temperature. As a rule, the density of the mat should vary by no more than 60 kg/m^3 (5 lb/ft^3) across the width of the mat. Surface temperature should not vary by more than 10° C (23° F) across the width of the mat.*

Each compactor in the compaction process, especially the initial phase compactor, must also be consistent in the approach to achieving consistently high density. Each compactor must work in such a manner that creates:

- Uniform pattern
- Uniform compaction force
- Uniform working speed
- Uniform temperature zone



Maintaining a consistent pattern and consistent density behind the paver can be challenging.

Some operators have difficulty repeating the same pattern as they follow the paver. They do not always hit each portion of the mat the same number of times. Therefore, density checks made by the quality control technician will vary. When this happens, the quality control technician or supervisor must work with the compactor operator to define the pattern and make sure that the pattern is being repeated.

Also, verify that the paving speed has not been changed. Often changes in paving speed are not communicated to the compaction team and the quality control team. A rolling pattern that has been working well is suddenly causing the compactor to fall behind the paver and to be working in a lower temperature zone, for example. The compactor falls behind because the paving speed has been increased. And, the operator tries to alter the rolling pattern to stay close to the paver.

Never change paving speed without doing two things. First, communicate the speed change to the compaction team. Second, verify that the initial phase compactor can keep up if the speed is being increased.

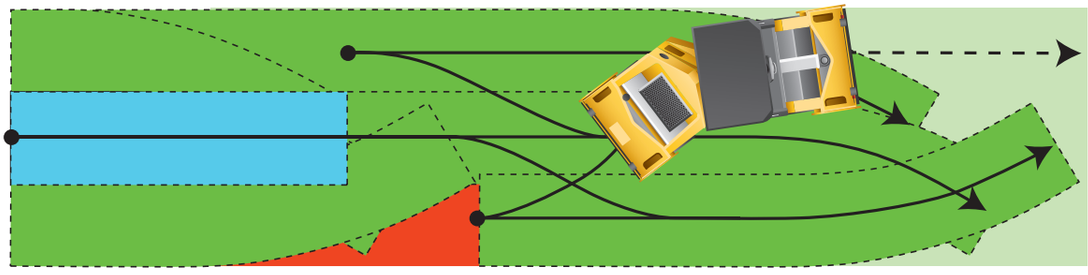




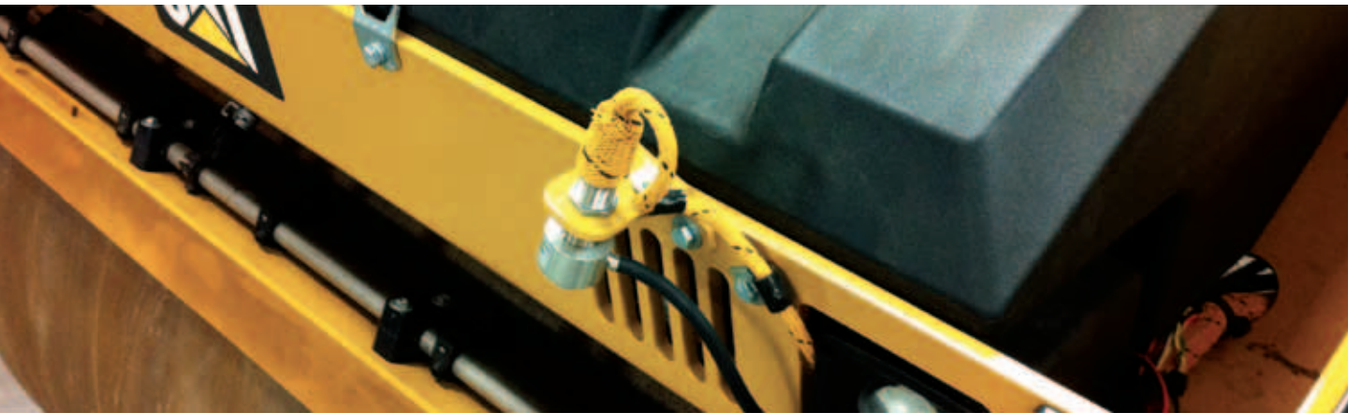
New technology helps operators maintain consistency of rolling patterns.

Options are available for asphalt compactors to help operators maintain uniform rolling patterns. Screens in the operator's compartment can be programmed to show the operator where the roller is located on the mat and how much of the pattern has been completed.

Global positioning systems provide very accurate maps of rolling patterns. The control can be programmed with the required number of passes. Then, the screen will display different colors as the passes are completed. The operator no longer has to guess at the end of the pattern for reversing. And, there is less chance that the operator will miss any of the areas in the pattern because the screen provides immediate feedback for quick corrective action.



Example of a pass-counting map.



Infrared sensors send temperature data to the display in the operator's compartment.

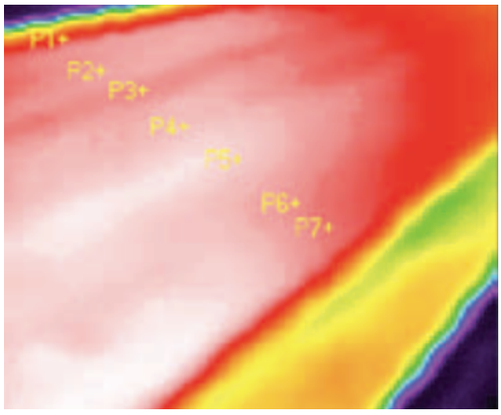
Infrared temperature sensors are another option for some asphalt compactors. On Cat models, sensors are installed on the front and rear of the machine. The sensors are continually cleaned by compressed air that keeps dust, fumes and moisture away from the sensor lenses. The temperature systems are accurate and provide constant, visual reference at the operator's station display. Not only does the operator know where the machine is located in relation to the defined rolling pattern, the operator knows where the machine is located relative to the desired temperature zone.

Large temperature variations are caused by long paver stops. The portion of the mat under the screed remains hot because it is confined. The portion of the mat just behind the screed loses heat because it is exposed to the elements.

Heat loss depends on mat thickness, air temperature and wind speed. If the mat temperature varies by more than 15° C (30° F), there is likely to be significant density variation. To help promote uniform density, limit paver stops to no more than five minutes.



Temperature variation caused by 10-minute paver stop. Screed imprint seen in infrared image is not visible in digital image.





Temperature variation caused by thin mat on the shoulder side.

In some instances, mat thickness varies across the width of the mat. The thinner portion will lose heat at a faster rate than the thicker portion. In the example shown above, the shoulder was higher than the driving lane in this area of the project. The mat laid over the driving lane was the specified 50 mm (2"). The mat thickness diminished to around 25 mm (1") over the shoulder. The density of the mat varied significantly due to temperature variation and also due to the thin mat having a low ratio of layer thickness to aggregate size. In this instance, all density checks taken in the shoulder failed to meet the minimum requirement, while all density checks taken in the driving lane passed the density requirement.

There is always a reason why density is variable. If you are troubleshooting density variability, look for variability in the paving process, mat temperature, rolling patterns, and rolling speed.

User Tip: *Temperature sensing and display are especially important when compacting mixes that have a tender zone. The operator can use the temperature display to verify that the initial compactor is staying ahead of the tender zone or that the intermediate compactor is staying behind the tender zone.*

Summary: *Compaction issues can be caused by a range of factors. Poor drum system maintenance, lack of planning, incorrect equipment selection, and inadequate operator training are just some of the factors that can cause problems during the compaction process. Certain mixes are more difficult to lay down and compact than others. In those cases, experimentation on the job may be the only solution when a mix is being used for the first time. What a crew learns from working on one project should be remembered and put to use on others when similar issues are encountered.*

GLOSSARY OF TERMS

- A -

Aggregate	Aggregate refers to the type or types of stones used in the manufacture of bituminous paving material.
Air Voids	Air voids are pockets of air trapped within an asphalt layer that has been laid down by a paving screed.
Ambient Conditions	Ambient conditions include air temperature, wind speed and direction, and cloud cover. Ambient conditions affect heat loss in the asphalt layer prior to final compaction.
Amplitude	Amplitude is the distance the drum moves into the asphalt layer. Amplitude is a critical measurement of impact force generated by a vibratory steel drum roller.
low	Drum movement between 0.25 – 0.5 mm (0.01 – 0.02")
mid	Drum movement between 0.5 – 0.75 mm (0.02 – 0.03")
high	Drum movement over 0.75 mm (0.03")
Asphalt	Asphalt is the generic term used to describe bituminous paving material, also called hot mix asphalt or sometimes warm mix asphalt.
Asphalt Cement	Asphalt cement is the oil used in the manufacture of bituminous paving material.
Asphalt Concrete	Asphalt concrete is another phrase used to describe bituminous paving material.
Asphalt Pick-up	Asphalt pick-up is the term used to describe hot asphalt sticking to the steel drum or to a rubber tire.
Axle Load	Axle load is the total weight applied by a steel drum or rubber tire.

- B -

Ballast	Ballast is weight added to or removed from a compactor in order to change the static force exerted by the compactor.
Base Layer	The base layer is usually the first layer of bituminous material in a roadway structure. The base layer usually consists of large aggregates and is laid 75 mm (3") thick or thicker.
Binder Layer	The binder layer is bituminous material laid on top of the base layer. The binder layer consists of medium size aggregates and is usually 50 - 100 mm (2 - 4") thick.
Bitumen	Bitumen is a black, sticky mixture of hydrocarbons found naturally or obtained as a residue from petroleum distillation. Bitumen is also called asphalt cement.
Bituminous Material	Bituminous material is the combination of aggregates, asphalt cement, and certain additives. Bituminous material is manufactured at asphalt plants.
hot mix	Bituminous material produced in an asphalt plant at temperatures between 149 – 177° C (300 – 350° F).
warm mix	Bituminous material produced in an asphalt plant at temperatures up to 38° C (100° F) lower than hot mix.
Bouncing Drum	Drum bouncing occurs when the asphalt layer cannot accept the amount of force being applied by a vibratory compactor. Drum bouncing is also referred to as de-coupling.

Breakdown Phase In some areas, the initial phase of compaction is referred to as the breakdown phase. The breakdown phase should accomplish the majority of the final target density.

- C -

Centrifugal Force Centrifugal force is an engineering calculation obtained by multiplying the mass of the eccentric weight by the radius of the weight rotation by the square of the speed of rotation (frequency). Centrifugal force has no direction relationship to compaction energy.

Cessation Temperature The cessation temperature is that temperature at which further attempts to gain density are likely to be unsuccessful. The cessation temperature varies by mix design, but as a rule, the cessation temperature is around 85° C (185° F).

Compaction Compaction is the mechanical process of reducing air voids and developing load-bearing strength in a layer of bituminous material by moving aggregates into interlocking contact.

Confined Edge A confined edge is an edge of an asphalt layer that is confined by a previously laid layer. The intersection of the two layers is called a longitudinal joint.

Cooling Curve A cooling curve is a graphical time chart of the heat loss in an asphalt layer based on layer thickness, type of material and ambient conditions.

Core Sample A core sample is a small portion of the cooled, compacted asphalt layer that is removed by quality control personnel and taken to a laboratory for quality analysis.

Core Temperature Core temperature refers to the temperature measured by a probe inserted into the center of an asphalt layer. Core temperature is always higher than surface temperature and is the true indicator of an asphalt layer's workability.

- D -

Dense-Graded Mix Dense-graded mixes are produced with a variety of aggregate sizes, asphalt cement and fines. The larger aggregates are surrounded by a mastic of fines and asphalt cement.

Density Density is the weight of a given volume of material, normally expressed as kilograms per cubic meter or pounds per cubic foot.

Density Testing Gauge Density testing gauges are used to test density on the project while the bituminous material is still hot enough to make adjustments. The quality control technician calibrates and uses the gauge on the project.

Density, Theoretical Maximum Theoretical maximum density is the weight of a given volume of bituminous material compacted in a repeatable, controlled manner in a laboratory environment.

Design Ratio Design ratio is defined as the ratio between the thickness of the asphalt layer and the largest aggregate in the layer. Most public works departments require at least a 3:1 ratio of layer thickness to aggregate size. The higher the design ratio, the easier is the compaction process.

- E -

Eccentric Weight The eccentric weight is an off-center mass inside the drum of a vibratory compactor. Rapid rotation of the eccentric weight creates forces that cause the drum to vibrate and to move into the asphalt layer.

Echelon Compaction Echelon compaction is a rolling pattern using two or more compactors for one phase of compaction.

Edge Cutter — An edge cutter is an attachment that can be installed on a compactor drum. An edge cutter trims the unconfined edge of an asphalt layer to provide a vertical face and straight line for joint matching.

Efficiency Factor — An efficiency factor should be used when calculating the working speed of a compactor to compensate for water refill stops and non-vibratory reversing. The normal efficiency factor for compactors is 75% to 85%.

End Gate — A paving screed has left and right end gates that confine the asphalt layer to the desired width. The bottom of an end gate has a metal strip called a ski. When the end gate is down, the ski floats on the grade and the edge of the asphalt layer is vertical for best joint matching.

- F -

Finish Phase — The finish phase of compaction is the final phase and has the goal of cleaning up any marks left in the surface by the previous compactors. Small gains in density may result from the finish phase.

Frequency — Frequency is the number of times that the drum hits the asphalt layer and is rated in vibrations per minute. Frequency can also be defined as the rotational speed of the eccentric weight shaft inside the drum.

- **low** — 40 – 46.7 HZ (2400 - 2800 vibrations per minute)
- **medium** — 46.7 – 56.7 Hz (2800 - 3400 vibrations per minute)
- **high** — Above 56.7 Hz (3400 vibrations per minute)

- G -

Gap-Graded Mix — Gap-graded mixes use an aggregate gradation with particles ranging from large to fine with some intermediate sizes missing. Gap-graded mixes are permeable with lots of stone-on-stone contact.

- I -

Impact — Impact is a dynamic compaction force. Impact occurs when the steel drum of a vibratory compactor moves into the asphalt layer.

Impact Marks — Impact marks are visible lines in the surface of the asphalt layer. Impact marks are caused by applying too much force (amplitude and weight) to the asphalt layer.

Impact Spacing — Impact spacing is the relationship between the frequency of a vibratory compactor and the working speed of a vibratory compactor. Impact spacing is the number of times per meter or foot that the drum moves into the mat.

Infrared Camera — An infrared camera produces an image of the temperature of the surface of the asphalt layer. The image usually encompasses the width of the mat for a distance of no more than 9 m (30').

Infrared Scanner — An infrared scanner is a hand-held device that measures and displays the temperature of a point at the surface of an asphalt layer.

Initial Phase Compaction — The initial phase of compaction occurs immediately behind the paver where the asphalt layer is the hottest. The initial phase should get the majority of the target final density.

Intermediate Phase Compaction — The intermediate phase of compaction occurs immediately after the initial phase in a temperature zone where the asphalt layer is still hot enough to allow for gains in density. The intermediate phase should achieve the final density target.

Internal Friction — Internal friction is the resistance to movement by the aggregate in an asphalt layer. The shape of the aggregate determines the amount of internal friction.

- L -

Leveling Layer — A leveling layer is a thin layer of bituminous material laid on a milled surface for the dual purpose of restoring profile and improving smoothness.

Longitudinal Joint — A longitudinal joint is the intersection of two asphalt layers along the edges that are parallel to the direction of paving.

hot/cold joint — A hot/cold longitudinal joint is created by laying down a hot asphalt layer adjacent to a cold, previously compacted asphalt layer.

hot/warm joint — A hot/warm longitudinal joint is created by laying down a hot asphalt layer adjacent to a layer laid down a short time before.

hot/hot joint — A hot/hot longitudinal joint is created by two pavers laying down adjacent mats in echelon.

- M -

Manipulation — Manipulation is a static force generated when forces exerted into the asphalt layer are not entirely vertical but are sent in all directions. Manipulation, normally associated with pneumatic compactors, helps seal the surface of the layer.

- N -

Notched Wedge Joint — A notched wedge joint is created by installing a joint shaper at the end of the paving screed. Notched wedge joints are often specified to avoid creating a large vertical unconfined edge that may be open to traffic.

- O -

Offset Drums — On some articulated dual drum compactors, the drums can be offset to increase the width of coverage beyond the standard drum width.

Open-Graded Mix — Open-graded bituminous material is made up of only a few aggregate sizes with intermediate sizes absent. Modified asphalt cement is usually part of the design. There is a lot of stone-on-stone contact.

Overhang — Overhang refers to the amount of drum that extends past the edge of the asphalt layer.

Overlap — Overlap refers to the amount of drum that extends onto the previously completed, adjacent pass.

- P -

Pass — A pass is one movement of the compactor in one direction from the starting point to the point where the rolling pattern ends or a new rolling pattern begins. In some locations, a pass is defined as movement forward and backward in the same coverage area.

Pass Mapping — Pass mapping through the use of a Global Positioning Satellite is an option on some compactors. A display at the operator's station shows the operator the machine's current position on the project and whether the compactor has successfully completed the designated rolling pattern.

Perpetual Pavement — Perpetual pavement is a term used to describe a full-depth asphalt structure designed to withstand an almost infinite number of axle loads without structural deterioration.

Polymer Modified Asphalt — A polymer is a synthetic compound in the form of a string of similar linked molecules. Polymers are added to asphalt cement to enhance layer strength at high temperature and layer elasticity at low temperature. Polymer-modified asphalt cement has high viscosity.

Profile — Profile is the cross section of a roadway structure, most specifically the slope of the surface of the roadway for drainage.

- Q -

Quality Anaysis — Quality analysis is testing, measuring and analyzing bituminous material and other specified aspects of a project in a laboratory or other controlled environment.

Quality Control — Quality control is testing, measuring and analyzing bituminous material and other specified aspects of a project at the jobsite while work is in progress.

- R -

Release Agent — A release agent is a liquid that helps prevent bituminous material from sticking to steel or rubber surfaces. Petroleum distillates, like diesel fuel, are prohibited in most locations because of their harmful effects on bituminous material. There are a variety of bio-degradable release agents.

Resonant Frequency — The resonant frequency is that combination of vibratory system frequency, amplitude, working speed, and asphalt layer stiffness that causes the drum to bounce (oscillate) away from the surface of the mat. A vibratory compactor should operate close to, but not at, resonant frequency.

Rolling Pattern — A rolling pattern consists of the number and order of passes with overlaps and overhangs that are required by the compactor to cover the width and length of its assigned area while matching the effective speed of the paving process.

- S -

Segregation — Segregation, in relation to asphalt paving, refers to the tendency of larger aggregates in a bituminous mixture to separate from the smaller particles and form pockets or stripes of large aggregates within the layer.

Specification, End Result — An end result specification is a written quality control / quality analysis series of measurement targets for items like ride quality, density and asphalt mix conformance.

Specification, Method — A method specification describes either the type of equipment or the technique(s) that must be used on a project.

Split Drum — A split drum asphalt compactor has drums that are split in the middle. Steering and propel speeds are synchronized to allow one half of the drum to turn faster or slower than the other half of the drum. Split drums are beneficial when compacting tight radius applications like cul-de-sacs.

Starter Boards — Starter boards are wood or metal strips placed under the paver screed before the screed is lowered to the starting point. Starter boards should be the same thickness as the compaction rate of the asphalt layer being laid down.

Static Pressure — Static pressure is axle load divided by the area of the drum or rubber tire that contacts the surface of the asphalt layer. Static pressure is expressed in kilopascals or pounds per square inch.

Static Linear Load — Static linear load is calculated by dividing axle load by the width of the drum. Static linear load is expressed as kilograms per centimeter or pounds per inch.

Stone Matrix Asphalt — Stone matrix asphalt is composed almost entirely of large aggregates, fines and modified asphalt cement. There is stone-on-stone contact, but the stones are coated with a thick mastic of fines and viscous asphalt cement.

- T -

Tack Tack is an emulsion consisting of paving grade oil, water and an emulsifying agent. Tack is applied to surfaces prior to paving to help improve the bond between layers.

Tamping Screed A tamping screed uses one or more tamper bars to deliver extra compaction energy to the asphalt layer and produce higher density in the asphalt layer prior to the compaction process. Tamping screeds typically also use vibration to help tighten surface texture.

Temperature Mapping Temperature mapping is an option on some asphalt compactors. Infrared temperature sensors send data to a display at the operator's station. The display tells the operator surface temperatures within the rolling pattern.

Tender Zone The tender zone is that temperature range in which the asphalt layer becomes tender and moves away from the drum rather than consolidating under the drum.

Test Strip A test strip is used to confirm that the proposed rolling pattern and equipment are adequate to achieve target density and match the paving production. The test strip may be part of the project or may be a separate element.

Transverse Joint A transverse joint is the perpendicular intersection of two layers of asphalt. Often a transverse joint is the resumption of paving from a cold, compacted layer. A transverse joint is also called a butt joint.

- U -

Unconfined Edge An unconfined edge is the edge of an asphalt layer that is open and not bound by an adjacent layer or gutter.

Utility Compactor Utility compactors have drum widths less than 1 m (40") wide and are typically used on low production projects or to supplement larger compactors on projects where there is a need for more maneuverable equipment.

- V -

Vibration Vibration is a dynamic compaction force. Vibration helps the aggregates in an asphalt layer to re-orient and move into closer contact. Vibration occurs when the eccentric weight inside the drum begins to rotate rapidly.

Vibratory Screed A vibratory screed delivers vibratory force to the asphalt layer as the layer passes under the screed. Screed vibration increases the density of the asphalt layer by a small amount and also helps tighten the surface texture.

Viscosity Viscosity refers to the flow rate of a liquid at a given temperature. The viscosity of asphalt cement used in bituminous material is affected by temperature and by the additives mixed with the asphalt cement. For compaction purposes, the higher the viscosity of the asphalt cement, the more difficult is the compaction process.

- W -

Water Distribution Mat Water distribution mats help spread water evenly to the surface of a compactor's steel drums.

Water Spray Nozzle Water spray nozzles are located on spray bars over the compactor's steel drums. Spray nozzles apply a fan-shaped spray pattern to the drum surface to help prevent hot asphalt from sticking to the drum.

Wear Layer The wear layer, or the friction or surface layer, is the final layer in the roadway structure. Usually the thinnest layer, it is designed to be the stiffest layer.

Wedge Joint A wedge joint is a tapered edge that eliminates a vertical face on the unconfined edge that may be open to traffic. A wedge joint is formed by installing a shaper at the end of a paver screed.

Wheel Covers Wheel covers are an option for pneumatic compactors. Wheel covers wrap around the front and rear axles to help the rubber tires maintain heat and minimize asphalt pick-up.

- Y -

Yield Yield refers to the linear distance that a given amount of bituminous material will pave at a certain depth and width.

